

Coupling-gated cortical coherence: what a field theory of conscious unity must assume, and what it predicts

Sebastian Bobadilla-Suarez
Independent Researcher
`sebastian.bobadilla.s@gmail.com`
ORCID: 0000-0002-5951-0772

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Abstract

A field account of cortical coordination holds that spatially distributed neural processes are coupled in part through the endogenous extracellular electromagnetic field. We ask what such an account must assume to reach a globally consistent, self-referential state, and what it predicts before it gets there. Its physical core is a noisy mean-field phase model whose coupling drifts through a synchronization threshold. If recovery of coherence after sleep is gated by that crossing, it is not free to take an arbitrary time course. For as long as the phase distribution tracks the stationary branch, it must show a square-root onset and a von Mises concentration–coherence relation with no fitted shape parameter. The tracking condition is part of the prediction, and the simulations measure its cost: ramp speed sets the apparent onset exponent, so an exponent fitted to a driven recovery trace reports the drive as much as the mechanism. A coupling plasticity that descends a squared-drift objective removes most of the objective available to it and keeps the system coherent, while driving within-cluster coupling far below between-cluster coupling—away from the

structure it is asked to learn, and further from it than from almost every relabelling of the nodes; a matched random update of the same step size does neither. Descending that functional does not produce representational learning, and its optimum is opposed to the target by a conservation law rather than by accident. A Lean 4 formalization composes finite capacity, Landauer’s bound, the noisy Kuramoto transition, sheaf gluing and Banach’s theorem into one conditional theorem, isolating the eight hypotheses it consumes and the two structures it takes alongside them, whose class fields carry physical content of their own; one witness assembled from distinct toy components satisfies all eight at once. We defend as a philosophical commitment, not a theorem, the identification of the glued section with the unity of a conscious episode and of its reflexive fixed point with a minimal self. Testing the account requires independent calibration of effective cortical coupling and an awakening experiment separating slow coupling change from alternative recovery mechanisms.

1 Introduction

Neural activity changes across sleep, awakening and anesthesia, but a change in synchrony does not by itself identify the physical interaction responsible for it. A field account makes a stronger proposal: spatially distributed neural processes are coordinated partly through the endogenous extracellular electromagnetic field, and changes in the medium can change that coordination (McFadden, 2002, 2020; Pockett, 2012). Ephaptic effects provide a possible interaction mechanism (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015). Whether that mechanism is sufficient to support the collective state such accounts require remains an empirical question.

Such an account has three requirements, and they order what follows. Coupling must be strong enough to sustain coherent collective activity; the

13 local descriptions that activity carries must be compatible enough to glue
14 into a single one; and that glued state must support an accurate model of
15 itself. The first requirement is physical and testable with current methods,
16 and most of this article is about testing it. The second and third are where
17 the mathematics is, and where the assumptions accumulate.

18 The first requirement admits a definite claim. If cortical coupling is slowly
19 gated—if recovery of coherence after sleep is driven by a coupling parameter
20 drifting through a synchronization threshold—then the recovery is not free
21 to take an arbitrary time course. For a recovery slow enough that the phase
22 distribution tracks the stationary branch, the theory fixes a square-root onset
23 and a one-parameter relation between phase concentration and coherence,
24 with no fitted shape parameter. That is a constraint an experiment can
25 meet or violate. How slow is slow enough is itself part of the claim, and
26 our simulations bound it from the other side, by showing which parts of the
27 constraint survive finite populations and finite ramp speeds and which do
28 not.

29 The contribution has four parts. The first is the coupling-gated recovery
30 hypothesis itself: a stationary coherent branch with square-root onset, a von
31 Mises concentration–coherence relation, and an experimental protocol with
32 failure conditions declared in advance (Sections 2 and 3). The second is a
33 constraint on how such data may be analyzed. The apparent onset expo-
34 nent measured from a finite ramp is a function of ramp speed across the
35 sweep of Section 4, so an exponent fitted to a recovery trace measures the

36 ramp at least as much as the mechanism—and that holds for any analysis
37 reading a critical exponent off a driven transition, not only for ours. The
38 third is a negative result about plasticity: a coupling update descending
39 a squared-drift objective removes most of the objective available to it and
40 preserves coherence, while moving *away* from the environmental structure
41 it is expected to learn—below almost every node relabelling, and beyond
42 what a matched random update of the same step size does. Descending a
43 squared-drift functional does not produce representational learning, and the
44 direction of that failure follows from a conservation law rather than from the
45 particular run. The fourth is a machine-checked composition (Section 5) in
46 which finite information capacity, Landauer’s bound, a predictive dissipation
47 bound, the noisy Kuramoto transition, sheaf gluing and Banach’s fixed-point
48 theorem yield one conditional theorem, with the eight hypotheses that com-
49 position consumes isolated and named; one witness assembled from distinct
50 toy components satisfies all eight at once, so their conjunction is not empty.

51 A prose table of assumptions would list the same eight items, so the fourth
52 contribution needs a word about method. What the Lean development adds
53 is that it makes the list non-negotiable. The characteristic failure in an
54 argument of this shape is not a false step but a silent change of subject:
55 a theorem about a stationary density read as a claim about a trajectory,
56 an inequality about a register read as a claim about learning, agreement of
57 phases read as compatibility of representations. Type-checking forbids each
58 of those substitutions, and an assumption needed to make a step go through

59 appears as a hypothesis in the term rather than as a word in a sentence. The
60 last of them is not merely forbidden but refuted by a constructed object: the
61 development carries a cover at a single common phase whose local densities
62 disagree where its patches meet, satisfying every other requirement of the
63 gluing theorem and failing compatibility alone (Section 5.2). That object
64 is what a prose table of assumptions cannot carry. Formalization does not
65 make the assumptions true; it makes them impossible to lose.

66 The physical model and the formal composition belong in one article be-
67 cause they are two halves of one commitment. The coupling-gated prediction
68 is worth testing under this framework, rather than as one more synchroniza-
69 tion model, only because the coherent state it produces is claimed to do
70 further work: to support a globally consistent representation and a reflexive
71 fixed point that we identify with phenomenal unity and a minimal self.

72 The distinction that does that work is between phases agreeing and de-
73 scriptions fitting together, and it is concrete. Take two cortical regions whose
74 receptive territories overlap, each carrying a probability distribution over
75 what occupies the shared part. Their oscillators can be locked to the same
76 phase at every instant while one region's distribution on the overlap places an
77 object left of fixation and the other places it right. The phases agree; the de-
78 scriptions do not, and no single distribution over the shared territory restricts
79 to both. A theory that identifies unity with synchrony cannot mark that dif-
80 ference, and so cannot separate a brain whose regions oscillate together from
81 one whose regions are about the same world. The sheaf condition marks

82 exactly it: local states glue to one global section when their restrictions to
 83 shared regions coincide, and otherwise not at all. Coherence enters as the
 84 physical condition under which compatibility can be maintained, not as a
 85 substitute for it, and the self-prediction map then asks the further ques-
 86 tion of whether the glued state contains an accurate model of itself. That
 87 identification is a commitment, not a theorem, and Section 6 states it, ar-
 88 gues for it, and says what would count against it. Its separation from the
 89 physical tests is what makes both assessable: the cortical coupling hypothe-
 90 sis and the recovery prediction can fail while the mathematical composition
 91 stands, and the composition can stand while the identification is wrong. The
 92 established ingredients (Landauer, 1961; Still et al., 2012; Sakaguchi, 1988;
 93 Banach, 1922) therefore appear here for their biological composition, not for
 94 any new fundamental physical law.

95 **2 A coupling-gated model of cortical coher-** 96 **ence**

97 **2.1 Dynamics and stationary solutions**

98 The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

Here $D > 0$ is phase diffusion, W_i are independent Wiener processes, and K_{ij} has inverse-time units, with angles treated as dimensionless. The uniform mean-field convention is $K_{ij} = K/N$. Frequency heterogeneity and phase diffusion are separate parameters; a frequency-spread width cannot be substituted for D when applying the identical-frequency stationary theory.

For identical frequencies in their rotating frame, the assumed stationary density is

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}. \quad (2)$$

The I_n are modified Bessel functions, and ψ is the collective phase. Lean verifies normalization, the density's circular order parameter, and the self-consistency equation. At $K \leq K_c = 2D$, the only nonnegative self-consistent order parameter is zero. At $K > K_c$, exactly one positive solution coexists with zero; the positive branch is continuous at threshold and strictly increasing with K .

The near-threshold expansion gives

$$\frac{r^2}{K - K_c} \longrightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

This is a result about the stationary solution branch. Of the coherent branch we prove existence, uniqueness, continuous emergence at threshold and strict growth in K , but not that it is dynamically selected. The dynamical mean-field limit—propagation of chaos—is the missing link between the finite sys-

tem and the stationary description (Sznitman, 1991), and no theorem in this development closes it. Exchange symmetry alone does not justify factorization of the joint law, particularly for an ensemble with an unpinned collective phase.

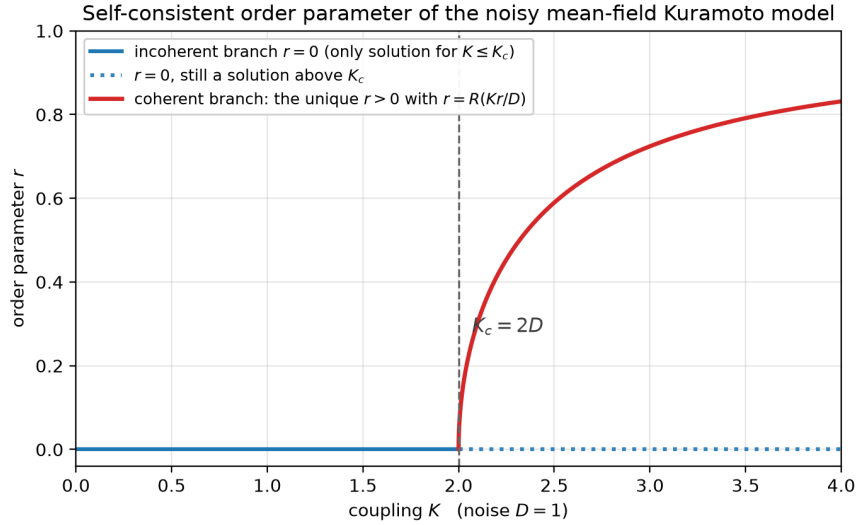


Figure 1: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the same integral expressions used in the formalization; it illustrates the static theorems and does not establish dynamical selection.

A spatial version replaces the finite sum by an integral against a kernel $K(x, y)$. We normalize the substrate measure to total mass one, so the mean-field strength is a kernel average, not a quantity that grows with the arbitrary measure of the substrate. Reduction of a heterogeneous spatial kernel to a scalar K is an additional modelling step, and the threshold in Eq. (2) does

126 not automatically apply to every spatial or signed network.

127 **2.2 Physical identification and calibration**

128 Regular mesh refinement gives convergence of a discrete coupling-energy
129 quadrature to its continuum integral: scalar energy convergence, not conver-
130 gence of trajectories and not a construction of a physical kernel. The model
131 posits a continuum kernel and separately identifies its cortical realization
132 with endogenous electromagnetic coupling. The formal predicate requires
133 a jointly continuous kernel, a normalized substrate, a mean-field strength
134 equal to its spatial average, and no single site providing more than half of
135 that mean-field coupling. Toy witnesses satisfy these conditions; none is a
136 model calibrated to cortex.

137 Extracellular fields can modulate spike timing (Fröhlich & McCormick,
138 2010; Anastassiou & Koch, 2015), and field-based models have been proposed
139 for integrating distributed neural information (Pinotsis & Miller, 2022; Mc-
140 Fadden, 2020). These observations motivate an effective interaction without
141 establishing its magnitude relative to phase diffusion, its spatial averaging
142 rule, or its causal role in experience; changes in phase organization under
143 propofol (Bastos et al., 2021) likewise do not measure a crossing of the model’s
144 coupling threshold.

145 The physical-unit conversion is the sharpest open limitation of the frame-
146 work. An estimate proportional to $N\Delta tf$, using population size, a spike-time
147 shift and oscillation frequency, is dimensionless. It cannot be compared di-

rectly with an inverse-time diffusion parameter. A conversion such as

$$K_{\text{eff}} = \gamma N s E f \quad (4)$$

requires a rate calibration γ , where s is the time shift per unit field and E the field amplitude. Supplemental Material sets out why γ is not recoverable from the other factors: it states how a per-cycle phase advance accumulates into a drift rate, and the choice that would let it cancel removes the very ratio the threshold test is about. The saved field-strength calculations are conditional on this calibration and on the chosen spatial aggregation; they do not independently establish $K_{\text{eff}} > 2D$ in cortex. Determining the interaction kernel and phase diffusion from the same physical system is necessary before the threshold becomes a quantitative biological claim; Section 3.2 makes that measurement the first item of the proposed experiment.

3 Recovery after awakening: a conditional prediction

Sleep-associated changes in extracellular space and clearance motivate a possible slow control of coupling (Xie et al., 2013). We hypothesize that recovery after awakening includes a change in effective coupling $K(t)$, mediated by extracellular geometry, on a slower timescale than phase dynamics. Neither the monotonic relation between geometry and coupling nor its role in hu-

man sleep inertia follows from the cited physiological observation; the model requires both to be measured.

If $K(t)$ rises through K_c and the phase distribution tracks the coherent stationary branch, Eq. (3) predicts

$$r(t) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad t > t_0, \quad \dot{K}(t_0) > 0. \quad (5)$$

The ideal stationary description has a subthreshold interval followed by continuous onset. Saturation requires a specified subsequent coupling trajectory; monotonic coupling alone does not determine a logistic or any unique sigmoid. The onset time is fixed by $K(t_0) = K_c$ only if $K(t)$ is independently known. When coupling is inferred from the same coherence trace, this is not a parameter-free prediction of latency.

Critical slowing down limits the adiabatic prediction of Eq. (5) near threshold, and a finite population carries nonzero fluctuating order below it. An observable onset therefore requires a predeclared threshold above the finite-size floor, and a finite-rate trajectory need not show an infinite-slope cusp. Section 4.1 quantifies the limitation.

3.1 Distributional and temporal tests

The stationary model makes a second prediction: phase concentration and coherence obey $r = I_1(a)/I_0(a)$, with no fitted shape parameter. Its applicability depends on the stationary-density assumption and the observation

185 model, and the relation is common to the whole von Mises family, so agree-
 186 ment with it identifies neither ephaptic coupling nor this framework against
 187 alternative synchronization models. Within the same ansatz $a/r = K/D$ for
 188 $r > 0$, so the limiting threshold ratio follows from the density relation instead
 189 of testing it, and ratios near the finite-size floor are especially unstable.

190 The estimators must be independent, which is the constraint most easily
 191 violated in practice. Estimating a by inverting the Bessel ratio from r would
 192 enforce the relation being tested, and linear statistics on wrapped phases
 193 estimate neither quantity.

194 **3.2 A protocol for the awakening experiment**

195 The requirements above define a design rather than a single measurement,
 196 and the design is where the content of the hypothesis lives. An adequate test
 197 needs all six of the following.

- 198 **1. Coupling and diffusion in the same units.** Estimate K and D
 199 from the same cortical regime, in inverse-time units, separating phase
 200 diffusion from quenched frequency spread and from observation noise.
 201 Without this the threshold $K > 2D$ is not an evaluable statement
 202 about cortex.
- 203 **2. Geometry calibrated to coupling.** A diffusion-MRI observable or
 204 an astrocytic signal is an indicator of extracellular geometry, not a mea-
 205 surement of K . Its mapping to effective coupling must be established

206 separately, with its own uncertainty.

207 **3. Phase estimation at adequate spatial resolution.** At least 100
208 spatial samples per estimate. Estimate r by the circular resultant
209 length and a by an unweighted regression of log-density on the co-
210 sine of phase relative to the circular mean. With fewer sites, pooling
211 over time changes the estimand to a spatiotemporal distribution and
212 must be stated as such.

213 **4. An independent recovery endpoint.** Behavioral recovery measured
214 separately from the phase trace, so that latency is not defined by the
215 signal whose latency is at issue.

216 **5. A predeclared onset criterion.** A coherence threshold fixed above
217 the finite-size floor before unblinding, chosen in the knowledge that a
218 finite ramp smears the cusp.

219 **6. Model comparison under matched observation.** Coupling-gated
220 recovery compared against exponential, flexible sigmoidal and other
221 slow-recovery models under identical temporal filtering and observation
222 noise. Latency alone cannot distinguish the mechanism, because many
223 processes create delays.

224 The design fails informatively in three ways: failure of the stationary
225 phase-density relation over a sufficiently broad concentration range; absence
226 of the proposed coupling-geometry association; or a recovery trace better

227 explained by an alternative after finite-rate and observation effects are ac-
 228 counted for. Two apparent failures are not failures. A linear apparent onset
 229 is evidence against the adiabatic approximation only where that approxima-
 230 tion is independently justified, and it does not falsify the field hypothesis
 231 under arbitrary ramp speeds. The absence of a fold in the stationary equa-
 232 tion excludes stationary coexistence of two positive branches in this model,
 233 but does not exclude finite-rate hysteresis or adjudicate theories of ignition.

234 **4 Simulation results**

235 The simulations characterize the finite-size, finite-rate regime that the sta-
 236 tionary theory leaves open, and they bear on whether the prediction of Sec-
 237 tion 3 is measurable at all. Two of them, Sections 4.1 and 4.3, illustrate
 238 methodological issues that also arise in analyses of driven transitions and
 239 dissipation-driven plasticity. Each is a finite run at its sampled parameters
 240 and not a general statement, and none discharges a hypothesis of the com-
 241 position in Section 5. Each run uses explicit seeds and compact saved sum-
 242 maries, and the numerical values below are generated from those artifacts.
 243 The supplement gives parameters, estimators, failure criteria and additional
 244 panels. Identical-frequency runs use phase diffusion rather than a quenched
 245 frequency-spread surrogate, and order plots display the $1/\sqrt{N}$ reference.

246 4.1 Ramp speed sets the apparent onset exponent

247 The dynamic-ramp study (Fig. 2) uses $N = 2000$, $D = 1$ and 32 replicas at
248 four coupling speeds. Escape is the first of three consecutive saved samples at
249 $r \geq 0.2$ after threshold. Only 5/32 replicas escape within the fastest window.
250 The three slower ramps have mean coupling delays 0.1947, 0.0684 and 0.0253,
251 with a fitted delay exponent 0.443. Their apparent onset exponents run from
252 0.898 to 0.417, against 0.443 to 0.448 returned by the same estimator on the
253 exact stationary branch sampled on the same coupling grids. At the fastest
254 speed the ensemble mean never reaches the declared fit window, so the value
255 the estimator returns there, 0.964, is a local log-slope of a delayed trajectory
256 rather than an onset exponent.

257 The exponent recovered by fitting a driven transition is a joint function
258 of the mechanism and the drive rate. An onset exponent read off a recovery
259 trace therefore says little about the underlying transition unless the ramp rate
260 is independently known and the finite-size floor is declared, and that holds
261 for any such analysis, not only for one conducted under this framework. The
262 prediction of Section 3 is the first thing it constrains: the stationary square-
263 root law cannot be imposed on an arbitrary recovery trace, and its absence
264 is not by itself evidence against a coupling-gated mechanism. The specified
265 deviation-based critical ramp speed is not bracketed by the sweep.

266 The dynamical-selection control shows escape toward a coherent branch
267 above threshold. At $K = 2.8$, steady mean order is 0.67735 against the static
268 value 0.68270. Refining the integration step by a factor of 4 at fixed dura-

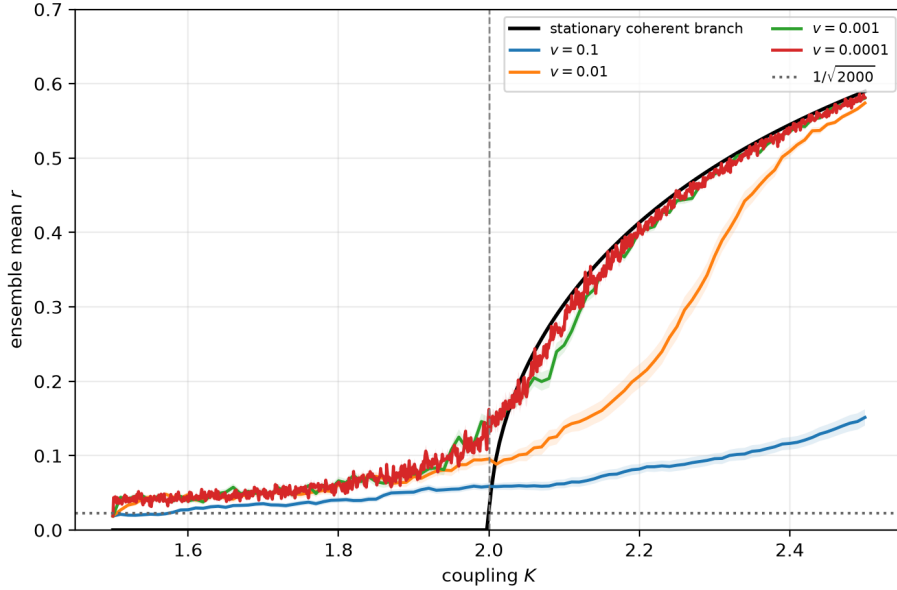


Figure 2: Finite-rate departures from the stationary branch. Ensemble-mean order is plotted against the increasing coupling for four ramp speeds v , with bands of one standard error across replicas. The dashed vertical line is the static threshold and the dotted horizontal line the finite-size reference. Faster ramps delay escape; the stationary branch is not a universal time course for recovery.

269 tion moves that order to 0.68168, so the bulk of the gap is Euler–Maruyama
 270 bias; the last refinement still moves it by 0.00147, so the remaining -0.00102
 271 bounds the finite-size part rather than measuring it. The fitted growth-rate
 272 slope is 0.41505 against the theoretical $1/2$, over a window bounded below
 273 by the finite-size floor and above by the static branch; the same integrator
 274 bias enters that fit. The propagation-of-chaos diagnostics show decreasing
 275 subcritical dependence with increasing population size. In the supercritical
 276 unpinned ensemble, however, pair correlation remains substantial because

277 realizations share a random collective orientation. This negative uncondi-
 278 tional result does not settle a conditional mean-field limit after removal of
 279 that orientation.

280 4.2 Spatial coupling, frustration and calibration

281 A periodic two-dimensional sheet with row-normalized exponential coupling
 282 remains coherent in the sampled empirical decay-length band under the de-
 283 clared total coupling and frequency spread. At decay lengths 0.1, 0.2 and
 284 0.3 mm, steady order is 0.9424, 0.9427 and 0.9427. This is one finite param-
 285 eter regime; fixing total coupling while changing range tests fragmentation,
 286 not whether cortex supplies that total coupling.

287 A row-balanced Dale-law network shows that row balance alone does not
 288 guarantee a floor-level incoherent baseline. In a separately specified weak-
 289 synapse regime, a uniform modulatory term produces an operational co-
 290 herence crossing inside the bracket $K_{\text{eff}}/D \in [1.857, 2.110]$, one step of a
 291 geometric coupling grid of ratio 1.136. That bracket contains the model's
 292 own mean-field threshold $K_c/D = 2$, so the run locates the crossing without
 293 separating it from the threshold. Signed coupling amplifies finite-size fluctu-
 294 ations in the same runs, whose incoherent baseline carries $N\langle r^2 \rangle > 1$, which
 295 is the direction a resolved crossing would be displaced in, and a finer grid
 296 is what would decide it. The quantity K_{eff}/D is dimensionless; carrying it
 297 over to cortex needs the rate calibration γ of Section 2.2, which neither this
 298 sweep nor the spatial control supplies.

299 4.3 Squared-drift plasticity does not learn structure

300 The predictive dissipation bound of Section 5 bounds nonpredictive memory
 301 by dissipation, and it is tempting to read a system that lowers its dissipation
 302 as thereby coming to mirror the structure of its input. The bound licenses no
 303 such reading: a memory carrying no information at all satisfies it with room
 304 to spare, so lowering dissipation presses toward less representation rather
 305 than toward better-aligned representation. This control exhibits a system in
 306 which the pressure is not merely toward less but away.

307 For fixed phases, consider the squared-drift objective

$$\sigma(K) = \frac{1}{D} \int \left[\omega(x) + \int K(x, y) \sin(\theta_y - \theta_x) d\mu(y) \right]^2 d\mu(x). \quad (6)$$

308 It is a quadratic functional of an affine image of the kernel. The formalization
 309 constructs the bounded drift operator A and its adjoint on the relevant L^2
 310 spaces, obtaining $\nabla\sigma = (2/D)A^*(AK + \omega)$, and the specified gradient flow
 311 decreases this objective at fixed phases. The objective is not generally a
 312 thermodynamic entropy-production rate, which requires probability-current
 313 information. Whether biological couplings follow this flow is an additional
 314 modelling question.

315 The joint control evolves noisy phases with clustered frequencies and
 316 nonzero grand-mean drive, while updating symmetric couplings every 50 fast
 317 steps. Couplings remain nonnegative, hollow and normalized to a fixed to-
 318 tal. The symmetric-edge derivative of $\frac{1}{2} \sum_i v_i^2$ is $(v_i - v_j) \sin(\theta_j - \theta_i)$, so

319 both endpoints contribute. Resource constraints and evolving phases re-
 320 move the fixed-phase monotonicity guarantee. Each seed runs three arms on
 321 one phase-noise stream: the gradient step, a symmetric random step of the
 322 same Frobenius norm, and a frozen kernel. Denying an implication requires
 323 exhibiting its antecedent, so the reported quantity is how much of the ob-
 324 jective the update actually removes: symmetric coupling cancels from the
 325 mean drift, which fixes a floor of 990 beneath the frozen arm’s 1380.6063 to
 326 1382.3448, and the gradient arm removes between 0.6773 and 0.6914 of that
 327 headroom. The random arm of matched step size removes none of it, so the
 328 descent belongs to the gradient. The random arm, however, ends an order
 329 of magnitude less deformed than the gradient arm: successive isotropic steps
 330 cancel, leaving cumulative norm growth of 1.2879 to 1.3056 against the gra-
 331 dient arm’s 12.4094 to 14.6958. Because the random arm barely enters the
 332 sparsification channel, it does not separate the gradient descent from cumu-
 333 lative deformation. The learning rate 0.2 is the rate at which the objective
 334 descends furthest across a sweep from 0.001 to 0.4 evaluated on one of the
 335 three reported seeds, not a rate at which the update is known to converge.

336 Across three seeds, second-half order lies between 0.9167 and 0.9222: co-
 337 herence survives the descent. Alignment with the cluster template is read
 338 from the ratio of mean within-cluster to mean between-cluster coupling,
 339 which is unchanged by rescaling the kernel. That ratio starts between 0.9824
 340 and 0.9948 and ends between 0.3507 and 0.6135, while the random arm leaves
 341 it at 0.9965 to 0.9997; against 2000 node relabellings the final kernel is fur-

342 ther from the true template than from a fraction 0.9510 to 1.0000 of them
 343 (Fig. 3). The reference for that fraction is the frozen arm on the same seeds,
 344 which reaches 0.6435 to 0.9110 without any update at all, and not a null
 345 median of one half; the increment the gradient step contributes is the differ-
 346 ence between those ranges, which at three seeds is small enough that they
 347 nearly meet. Two readouts do not carry the effect and we do not rest it
 348 on them. The initial-minus-final Frobenius distance to the template lies be-
 349 tween -58.0962 and -48.1760 , so that distance grows; but the kernel norm
 350 grows by a factor of 12.4094 to 14.6958 over the same interval, and a fixed
 351 total resource constrains the sum of the entries rather than the norm, so
 352 most of that growth is concentration of mass onto fewer edges rather than
 353 loss of alignment. The edge-level correlation with the template ends between
 354 -0.0362 and -0.0208 , near its starting value: at this sparsity the edge-level
 355 variance swamps a contrast that the block means resolve. The template is a
 356 covariance of shared latent cluster signals, not an input to the update or a
 357 measured covariance of the constant frequencies.

358 The objective descends by most of what is available to it, coherence is
 359 maintained, and within-cluster coupling falls to a fraction of between-cluster
 360 coupling. Coherence is therefore not a proxy for representational alignment
 361 in this regime, and a plateau in the objective establishes neither stationar-
 362 ity of the adaptive process nor a thermodynamic entropy-production rate.
 363 Nor is the direction of the failure accidental. Symmetric coupling conserves
 364 the mean drift, so minimizing $\sum_i v_i^2$ can only equalize drift across nodes,

365 and equalizing drift rewards coupling between clusters of unlike frequency—
 366 the complement of a template that rewards coupling within clusters of like
 367 frequency. The objective and the template are in this pairing opposed by con-
 368 struction, and that is the point: nothing about a system’s dissipation fixes
 369 which structure its slow variables acquire. What fails here is the objective
 370 rather than thermodynamically specified learning at large: an Ising model
 371 whose couplings are annealed degrees of freedom, driven by external fields
 372 that carry the data, memorizes and generalizes under ordinary statistical-
 373 mechanical rules (Corberi et al., 2026). Those couplings equilibrate under
 374 a Hamiltonian containing the data; these descend a squared-drift functional
 375 whose stationary point the data never enters. Which functional the slow
 376 variables follow is what decides whether they acquire environmental struc-
 377 ture.

378 **4.4 The estimation protocol demonstrated on public** 379 **EEG**

380 Item 3 of Section 3.2 specifies estimators whose behavior on real recordings
 381 is not obvious, so we exercise them on public propofol-sedation EEG (Bajwa
 382 et al., 2025). The purpose is to show what the protocol produces and where
 383 it is fragile, not to test the awakening hypothesis, for which these data are
 384 the wrong design.

385 Two fragilities appear immediately. The implemented estimator pools 62

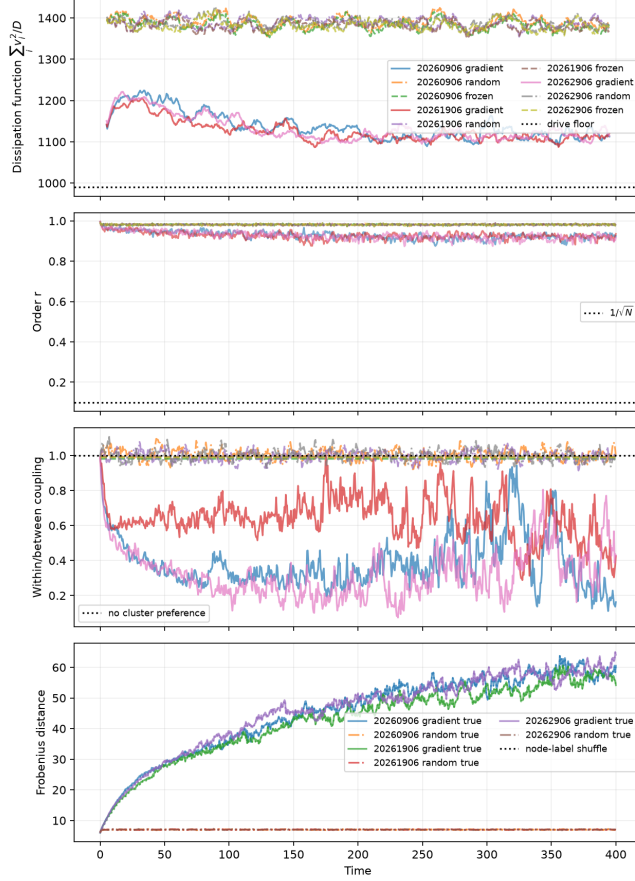


Figure 3: Joint fast/slow dynamics, three arms per seed on one phase-noise stream: gradient (solid), matched-norm random step (dash-dotted), frozen kernel (dashed). First: smoothed squared-drift dissipation against the drive floor the conserved mean drift imposes; only the gradient arm descends. Second: order and the finite-size reference. Third: mean within-cluster over mean between-cluster coupling, a measure unchanged by rescaling the kernel; the gradient arm falls well below the no-preference line while the other arms stay on it. Fourth: Frobenius distance to the true template (solid) and to a node-label shuffle (dotted), which track each other because the kernel norm, not the alignment, is what grows. Coherence persists while alignment is lost. The result limits the proposed connection between dissipation minimization and environmental structure.

channels over a 100-ms window, so it estimates a spatiotemporal distribution rather than an instantaneous spatial one—a departure from the protocol forced by channel count, and one that any scalp implementation will face. Bipolar and common-reference montages give different residuals, making referencing a material analysis choice rather than a preprocessing detail.

The bipolar results are compatible with the Bessel relation over the observed concentration range. That range is too narrow to distinguish the curve from its linear approximation at the observed scatter, so the check is uninformative about the relation itself. An informative test needs a broader concentration range and explicit treatment of source mixing, finite samples and temporal pooling; intracranial or high-density recordings across a genuine state transition are the natural setting.

5 Thermodynamic constraints and the formal composition

The synchronization model of Section 2 is one link of a longer conditional construction, whose nine nodes are:

n1 a finite phase space has bounded information capacity; **n2** a broken symmetry forces a boundary, where the field leaves its vacuum manifold; **n3** the boundary erases, and erasure costs heat; **n4** that dissipation bounds nonpredictive memory; **n5** discrete couplings coarse-grain to a continuum limit; **n6** a specified physical field realizes that limiting coupling; **n7** above threshold the field carries a unique coherent order parameter; **n8** compatible

409 local states glue to a unique global section; **n9** that section is the
410 unique fixed point of a self-prediction map.

411 Node-level results are established or cited; what the composition consumes
412 are the eight implications between them. Figure 4 shows where they sit and
413 what kind of assumption each one is, and Table 1 states them, classified by
414 their role in the composition rather than by their plausibility. In particular,
415 finite capacity does not imply a symmetry-breaking boundary: n1 and n2 are
416 independent roots, even though the composition can accept an implication
417 between them as an argument.

418 **5.1 Capacity, boundaries and predictive memory**

419 For a finite state space X , Shannon entropy satisfies $H(p) \leq \log |X|$, with
420 equality only for the uniform distribution. A stream with more possible
421 histories than $|X|$ cannot be encoded injectively in that register. Separately,
422 a continuous field on a connected substrate that reaches distinct components
423 of a disconnected vacuum manifold must leave that manifold somewhere,
424 the simplest topological-defect construction (Kibble, 1976). Relating this
425 locus to a particular finite information-processing register is the additional
426 hypothesis E12 carries.

427 A non-surjective update on a finite register is non-injective. Under the
428 specified system–bath model and thermal assumptions, erasure carries a Lan-
429 dauer heat cost (Landauer, 1961). Finite capacity does not imply that every
430 update erases, and logical many-to-one processing must be distinguished from

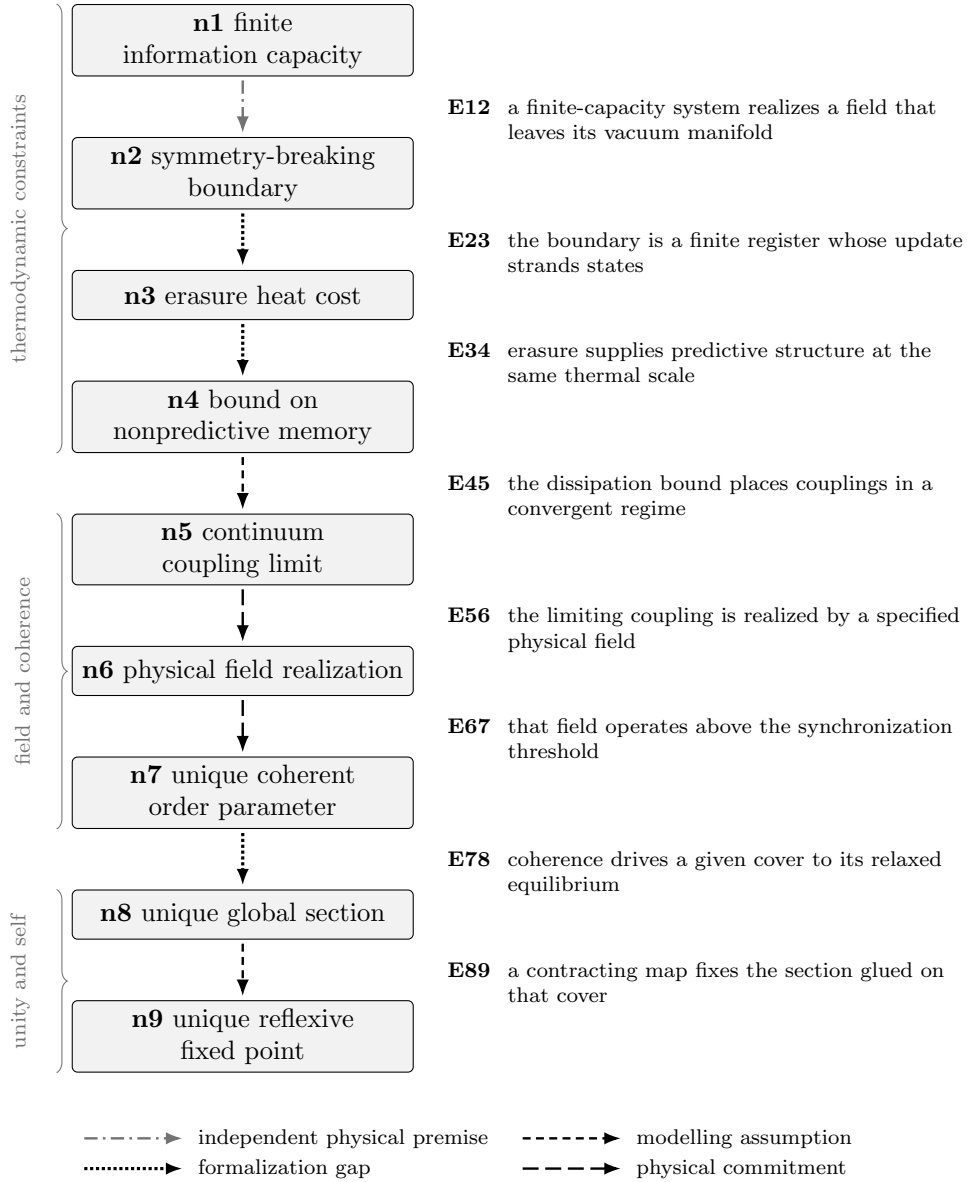


Figure 4: The conditional composition. Each node is an established or cited result; each arrow is an implication the composition consumes and no theorem here supplies, styled by the kind of assumption it carries. The three formalization gaps bracket the two physical commitments, so the chain’s empirical content sits between links of a different kind on either side. E12 is drawn as a premise because n1 and n2 are independent roots. Table 1 states each connection in full.

Edge	Status	Required connection
E12 capacity $\rightarrow$ boundary	Independent physical premise	A finite-capacity system realizes a field that leaves its vacuum manifold. Capacity alone does not imply this.
E23 boundary $\rightarrow$ erasure	Formalization gap	The boundary is associated with a finite register whose update leaves states unreachable.
E34 erasure $\rightarrow$ prediction	Formalization gap	Erasures supplies a predictive structure with the same thermal scale and heat budget, and nonzero wasted memory.
E45 prediction $\rightarrow$ continuum	Modelling assumption	A predictive dissipation bound places coupling energies in a convergent coarse-graining regime.
E56 continuum $\rightarrow$ field	Physical commitment	The limiting coupling is realized by a specified field satisfying the kernel conditions. Its identification with cortex is empirical.
E67 field $\rightarrow$ coherence	Physical commitment	That field operates above the synchronization threshold.
E78 coherence $\rightarrow$ unity	Formalization gap	Coherence drives a cover, supplied as a separate argument of the theorem, to an equilibrium reached by relaxation at coupling at least the named mean-field value. That its patches carry compatible local states is a field of the cover rather than a conjunct of this edge.
E89 unity $\rightarrow$ self	Modelling assumption	The state glued on that same cover is fixed by a contracting map mediated by its local self-encoding.

Table 1: The assumptions consumed by the conditional composition theorem, each an implication between two of the nine nodes above. Node-level results do not establish these connecting implications. The complete theorem map and witnesses are in Supplemental Material, Table S1.

431 its physical implementation (Norton, 2005; Shenker, 2000). The formaliza-
 432 tion records the physical heat identity as an obligation of each model and
 433 supplies an explicit one-bit eraser satisfying it.

434 For a system state X_t driven by an environmental signal S_t , the predictive
 435 dissipation bound is

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (7)$$

436 Under the no-back-action Markov structure $X_t \rightarrow S_t \rightarrow S_{t+1}$, data process-
 437 ing makes the bracket nonnegative (Still et al., 2012). The bound constrains
 438 nonpredictive memory at a given dissipation budget. It does not specify
 439 a learning rule, maximize prediction, or require the internal representation
 440 to reproduce environmental geometry—the point the plasticity control of
 441 Section 4 makes concretely. Connecting this budget to Landauer heat ad-
 442 ditionally requires nonpredictive information to be bounded by the entropy
 443 erased by the register. The formal bridge derives Eq. (7) from Landauer’s
 444 bound under that identification; it does not derive the identification.

445 A bound $D_{KL} \leq \Delta t \sigma$ also cannot force $D_{KL} \rightarrow 0$ unless its right-hand
 446 side tends to zero, and Gibbs’ inequality proves nonnegativity of relative en-
 447 tropy, not a general identification of relative entropy with physical entropy
 448 production. The chain uses a different technical sense of information at each
 449 step—a counting bound on a register, a thermodynamic heat cost, a predic-
 450 tive mutual information—and those senses are not interchangeable (Rosas

et al., 2026). These distinctions prevent an informational inequality from being read as a derivation of environmental learning, Bayesian inference, or semantic content.

5.2 Compatible states and reflexive fixed points

A sheaf formalizes local descriptions and their restriction to smaller regions. If local sections agree on every overlap, the sheaf condition gives a unique global section. In the probability-sheaf construction, this is consistency of local density information, and it is not implied by agreement of oscillator phases: the development includes a counterexample with equal phases but incompatible densities on an overlap. Section 6 rests on this distinction, and the model excludes such configurations only through its compatibility hypothesis.

A thermodynamic cover couples this gluing construction to a phase configuration at a potential minimum. For the deterministic positive-coupling model, convergence to that minimum is proved under an explicit initial basin condition; other equilibria prevent an unconditional conclusion. The stationary noisy order parameter, the deterministic relaxation, and the cover must therefore be connected by E78 rather than identified without a hypothesis. Neither field continuity nor synchronization alone guarantees a globally compatible representation.

A reflexive extension equips global sections with a complete metric and a predictive map formed by restricting a state to a local encoding region and

473 reading that encoding out. If the map contracts and fixes the section supplied
474 by the cover, Banach’s theorem gives uniqueness of that same fixed point.
475 The contraction rate is assumed, and the equality between the glued section
476 and the fixed point is part of E89. A coherent order parameter supplies the
477 supercritical inequality required by the stated contraction bound; it does not
478 construct the map or prove that cortical dynamics implements it.

479 The theorem `chain` concludes `UnifiedSelf` from these hypotheses and
480 from two structures it also takes as arguments, a thermodynamic cover and
481 a reflexive boundary. The eight edges are not the whole of what is assumed:
482 the cover’s overlap-compatibility field—the load-bearing assumption of the
483 unity-is-not-synchrony argument—and the boundary’s resonance condition
484 are class fields, so they are consumed by the theorem without appearing
485 among its edges. A single witness satisfies all eight at once, so their conjunc-
486 tion is not empty; it is assembled from a double well, a register, a uniform-grid
487 mesh and a three-site cortex, which remain distinct toy systems rather than
488 one mechanism. The project declares no additional Lean axioms beyond
489 the foundational axioms used by Lean’s libraries; physical content resides
490 in hypotheses and model fields. This checks the conditional mathematical
491 composition, not the empirical truth of its premises. Detailed constructions,
492 scope statements and counterexamples are retained in the supplement and
493 source files.

494 6 What the construction commits to about 495 consciousness

496 The composition terminates in a mathematical object, and we take a posi-
497 tion on what that object is. The commitment has two clauses. *Unity*: the
498 unique global section glued from compatible local states is the formal corre-
499 late of the unity of a conscious episode—the fact that a subject undergoes
500 one experience rather than a collection of co-occurring local ones. *Selfhood*:
501 the fixed point of the self-prediction map on that same section is the formal
502 correlate of a minimal self, a state that is what it takes itself to be. Neither
503 clause is a theorem, and no simulation here supports either. We argue for
504 them, then say what would count against them.

505 Three features of the construction motivate the identification, and each
506 distinguishes it from a bare synchrony measure.

507 First, compatibility rather than agreement. Oscillator phases can agree
508 everywhere while the local probabilistic descriptions they carry remain in-
509 compatible on their overlaps, and Section 5.2 exhibits that configuration
510 inside the formalization, not only as an illustration. If the unity of experi-
511 ence is a matter of local contents being about one world rather than merely
512 co-timed, then compatibility, not coherence, is the property to look for—and
513 coherence enters as the physical condition under which compatibility can be
514 maintained, which is what E78 asserts.

515 Second, uniqueness follows from the construction rather than from a sep-

516 arate selection principle. Accounts of conscious unity need something that
 517 picks one experience from among the candidates a system could support;
 518 integrated information theory postulates exclusion for this purpose (Tononi
 519 et al., 2016). Here uniqueness arrives twice as a consequence: given a cover
 520 whose local sections agree, the sheaf condition gives exactly one global sec-
 521 tion, and given a contracting self-prediction map, Banach’s theorem gives
 522 exactly one fixed point. The scope of that claim needs stating exactly, be-
 523 cause the stronger reading is available and is not ours. Uniqueness is relative
 524 to a cover and a map, and choosing them—which regions, which overlaps,
 525 which encoding region, and so where the subject ends—is not supplied by ei-
 526 ther theorem. It is what E78 and E89 carry. What the construction removes
 527 is the need for an additional principle to select among the experiences a fixed
 528 cover and map would otherwise admit; it does not remove the need to say
 529 which cover and map a given subject has. That is a structural argument,
 530 not evidence.

531 Third, the reflexive clause makes the self a condition rather than an addi-
 532 tion. The predictive map restricts a global state to a region encoding it and
 533 reads that encoding back out, so a fixed point is a state whose self-model is
 534 accurate about the state it models. That condition is available only because
 535 the state modelled is a single glued object: a system whose local descrip-
 536 tions do not cohere has no such state for a self-model to be about. Unity
 537 and selfhood are therefore one construction read at two levels, which is why
 538 `UnifiedSelf` requires the fixed point to be the glued section itself rather

539 than an unrelated section.

540 The costs belong with the argument. The commitment does not address
541 why any physical condition is accompanied by experience; it is a proposal
542 about which physical state is the conscious one, not about why states of that
543 kind feel like anything. It is therefore open to the objection that a system
544 could satisfy every condition—compatible local states, a supercritical field,
545 an accurate self-model—while nothing is experienced. We have no reply to
546 that objection and claim none. The identification is also multiply realizable
547 as stated: nothing in the sheaf or fixed-point conditions is specific to cortex
548 or to electromagnetic coupling, so the substrate claim is carried entirely by
549 E56 and E67 and is empirical. No conclusion about artificial systems follows
550 from the formal results.

551 What could count against the commitment is dissociation. It predicts that
552 the two clauses and the presence of a unified conscious episode go together,
553 so a state in which local cortical descriptions are demonstrably compatible
554 and the field supercritical, while the subject reports no unified experience,
555 would tell against it; so would unified reported experience where compati-
556 bility demonstrably fails. Anesthetic emergence, dreaming and dissociative
557 states are the places to look, because they separate coherence from report
558 in ways ordinary waking does not. Such tests need an operational measure
559 of overlap compatibility on neural data, which this article does not supply;
560 constructing one is the main obstacle to making the commitment empirically
561 live rather than merely stated. Supplemental Material specifies what such an

562 observable would have to satisfy—a restriction map, a statistic with a null
563 under decoding noise, validation against configurations whose compatibility
564 is built rather than inferred—and the three ways it would fail silently, one
565 of which is reading compatibility off synchrony and so assuming away the
566 distinction being tested.

567 The physical tests of Sections 3 and 4 are independent of all of this. They
568 concern coupling, coherence and recovery, and can fail while the identification
569 stands, or succeed while it is wrong.

570 **7 Discussion**

571 The framework connects thermodynamic information constraints, collective
572 phase organization and compatible local representations, conditionally and
573 with its assumptions named. Its biological content is a coupling-gated re-
574 covery hypothesis with defined observables, a stated protocol, and failure
575 conditions declared in advance. Its formal content is a composition whose
576 eight gaps are exhibited rather than argued past, and which can be inhabited
577 by a single model. Its interpretive content is the identification of Section 6,
578 offered as a commitment with reasons.

579 The simulations are useful because they restrict the hypothesis rather
580 than support it. Finite ramps smear the stationary onset to the point where
581 a fitted onset exponent chiefly reports the ramp; an unpinned collective
582 phase makes independence diagnostics ambiguous in exactly the supercritical

583 regime the model cares about; and squared-drift plasticity maintains coher-
584 ence while degrading structural alignment. The second and third of these
585 are results about method and about a common intuition, and they transfer
586 beyond this framework.

587 The proposal differs in emphasis from accounts based on integrated in-
588 formation (Tononi et al., 2016), global workspace dynamics (Baars, 1997;
589 Dehaene et al., 2014), or predictive processing (Friston, 2010). It specifies a
590 candidate interaction medium and a mathematical compatibility condition,
591 and, relative to a given cover and self-prediction map, it derives the unique-
592 ness those accounts postulate. It does not derive the exclusion principle of
593 integrated information, identify every broadcast with a field transition, or
594 derive a generative model from a dissipation bound. Continuous fields can
595 carry incompatible local descriptions and discrete models can admit sheaf
596 constructions, so continuity alone does not settle the combination problem;
597 compatibility is what this account offers in its place.

598 Two formal results bear on the substrate question, and both admit a
599 stronger reading than they support. At matched coupling resource, a fixed
600 wiring support that excludes a better-correlated pair can be suboptimal—a
601 statement about constrained optimization, not about biology. On an atom-
602 less substrate, a kernel supported on finitely many points contributes zero to
603 the chosen continuum coupling integral. That is a property of the measure
604 and the observable, and it cuts against this framework’s own coarse-graining
605 step as much as against anything else. It does not imply that digital hard-

606 ware cannot approximate a field or support consciousness, and no biological-
607 versus-artificial conclusion follows.

608 The empirical priorities follow directly from Section 3.2: estimate effec-
609 tive coupling and diffusion in consistent units, test the proposed relation
610 between coupling and extracellular geometry, and measure recovery against
611 an independently defined behavioral endpoint under matched model compar-
612 ison. A favorable concentration–coherence relation would support a statis-
613 tical description, not identify its physical cause; an unfavorable result in a
614 regime satisfying the stated assumptions would constrain that description.
615 Successful synchronization simulations cannot substitute for measurements
616 distinguishing ephaptic interactions from other coupling mechanisms.

617 The article therefore asks to be judged in three separable parts: a pre-
618 diction that can be tested, a composition that can be checked, and an iden-
619 tification that can be argued about. Keeping them separate is what makes
620 the whole assessable.

621 **Data and code availability**

622 The Lean development, simulation scripts, tests and saved numerical sum-
623 maries are maintained in the manuscript source repository. Publication sim-
624 ulation values are regenerated from saved summaries without rerunning pro-
625 duction sweeps. The exploratory EEG analysis uses OpenNeuro dataset
626 ds005620 (Bajwa et al., 2025); preprocessing and selection are described

627 in Supplemental Material. The source repository contains the reproducible
628 analysis and does not redistribute the raw recordings.

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630 This research received no external funding. The author declares no compet-
631 ing interests.

632 **Use of AI tools**

633 AI assistance to this work was substantial and is not separable by task.
634 Several coding assistants and chat interfaces were used in an interleaved way
635 throughout: Google’s Antigravity CLI (Gemini 3.1, Claude 4.6 Opus, Claude
636 4.6 Sonnet), OpenAI’s Codex CLI (Astra, Terra, Luna, Sol), Claude Code
637 (Claude Opus 5), Gemini Pro through its web interface, and Fable (Claude
638 Opus 4) through OpenRouter, accessed with the Hermes CLI. Between them
639 they contributed to the Lean development, the simulation and analysis code,
640 and the drafting and revision of both documents. Because the tools were
641 used in combination and in alternation, no section, proof, figure or result is
642 attributable to any one of them, and none is claimed to be. The author is
643 responsible for all content, including everything a tool produced. Machine
644 checking establishes the stated Lean results under their hypotheses; it does
645 not validate biological interpretations or replace scientific review of generated

code and prose.

References

- Pinotsis, D. A., & Miller, E. K. (2022). Beyond dimension reduction: Stable electric fields emerge from and allow representational drift. *NeuroImage*, 253, 119058.
- Tononi, G., Boly, M., Massimini, M., & Koch, C. (2016). Integrated information theory: from consciousness to its physical substrate. *Nature Reviews Neuroscience*, 17(7), 450-461.
- Baars, B. J. (1997). In the Theater of Consciousness: The Workspace of the Mind. *Oxford University Press*.
- Dehaene, S., Charles, L., King, J. R., & Marti, S. (2014). Toward a computational theory of conscious processing. *Current Opinion in Neurobiology*, 25, 76-84.
- Friston, K. (2010). The free-energy principle: a unified brain theory?. *Nature Reviews Neuroscience*, 11(2), 127-138.
- Pinotsis, D. A., & Miller, E. K. (2023). In vivo ephaptic coupling allows memory network formation. *Cerebral Cortex*, 33(17), 9877-9890.
- Pinotsis, D. A., Fridman, G., & Miller, E. K. (2023). Cytoelectric coupling:

664 Electric fields sculpt neural activity and "tune" the brain's infrastructure.
665 *Progress in Neurobiology*, 226, 102465.

666 McFadden, J. (2002). The conscious electromagnetic information (cemi) field
667 theory: the hard problem made easy?. *Journal of Consciousness Studies*,
668 9(8), 45-60.

669 Pockett, S. (2012). The electromagnetic field theory of consciousness: a
670 testable hypothesis about the characteristics of conscious as opposed to
671 non-conscious fields. *Journal of Consciousness Studies*, 19(11-12), 191-223.

672 McFadden, J. (2020). Integrating information in the brain's EM field: the
673 cemi field theory of consciousness. *Neuroscience of Consciousness*, 2020(1),
674 niaa016.

675 Fröhlich, F., & McCormick, D. A. (2010). Endogenous electric fields may
676 guide neocortical network activity. *Neuron*, 67(1), 129-143.

677 Anastassiou, C. A., & Koch, C. (2015). Ephaptic coupling to endogenous
678 electric field activity: why bother?. *Current Opinion in Neurobiology*, 31,
679 95-103.

680 Pockett, S. (2002). Difficulties with the electromagnetic field theory of con-
681 sciousness. *Journal of Consciousness Studies*, 9(4), 51-56.

682 Vöröslakos, M., et al. (2018). Direct effects of transcranial electric stimulation
683 on brain circuits in rats and humans. *Nature Communications*, 9, 483.

684 Verkhatsky, A., & Nedergaard, M. (2018). Physiology of Astroglia. *Physio-*
685 *logical Reviews*, 98(1), 239-389.

686 Bastos, A. M., Donoghue, J. A., Brincat, S. L., Mahnke, M., Yanar, J., Cor-
687 rea, J., ... & Miller, E. K. (2021). Neural effects of propofol-induced uncon-
688 sciousness and its reversal using thalamic stimulation. *eLife*, 10, e60824.

689 Bajwa, I. J., Nilsen, A. S., Skukies, R., Aamodt, A., Ernst, G., Storm, J.
690 F., & Juel, B. E. (2025). A repeated awakening study exploring the capac-
691 ity of complexity measures to capture dreaming during propofol sedation.
692 *Scientific Reports*, 15, 32012.

693 Xie, L., Kang, H., Xu, Q., Chen, M. J., Liao, Y., Thiagarajan, M., ... &
694 Nedergaard, M. (2013). Sleep drives metabolite clearance from the adult
695 brain. *Science*, 342(6156), 373-377.

696 Norton, J. D. (2005). Eaters of the lotus: Landauer's principle and the return
697 of Maxwell's demon. *Studies in History and Philosophy of Science Part B:*
698 *Studies in History and Philosophy of Modern Physics*, 36(2), 375-411.

699 Landauer, R. (1961). Irreversibility and heat generation in the computing
700 process. *IBM Journal of Research and Development*, 5(3), 183-191.

701 Kibble, T. W. B. (1976). Topology of cosmic domains and strings. *Journal*
702 *of Physics A: Mathematical and General*, 9(8), 1387-1398.

703 Sakaguchi, H. (1988). Cooperative phenomena in coupled oscillator systems
704 under external fields. *Progress of Theoretical Physics*, 79(1), 39-46.

705 Still, S., Sivak, D. A., Bell, A. J., & Crooks, G. E. (2012). Thermodynamics
706 of prediction. *Physical Review Letters*, 109(12), 120604.

707 Kawai, R., Parrondo, J. M. R., & Van den Broeck, C. (2007). Dissipation:
708 The phase-space perspective. *Physical Review Letters*, 98(8), 080602.

709 Corberi, F., dello Russo, S., Messuti, G., Scarpetta, S., & Smaldone, L.
710 (2026). Thermodynamic learning. *arXiv preprint*. arXiv:2609.04732.

711 Banach, S. (1922). Sur les opérations dans les ensembles abstraits et leur ap-
712 plication aux équations intégrales. *Fundamenta Mathematicae*, 3(1), 133-
713 181.

714 Shenker, O. R. (2000). Logic and entropy. *PhilSci-Archive preprint*. philsci-
715 archive.pitt.edu/115.

716 Rosas, F. E., Mediano, P. A. M., Luppi, A. I., Boyd, A. B., Berta, M., Jensen,
717 H. J., Gastpar, M., & Carroll, S. M. (2026). Four faces of information in
718 natural systems. *Nature Reviews Physics*. doi:10.1038/s42254-026-00971-4.

719 Sznitman, A.-S. (1991). Topics in propagation of chaos. In *École d'Été de*
720 *Probabilités de Saint-Flour XIX—1989*, Lecture Notes in Mathematics,
721 vol. 1464, pp. 165–251. Springer. doi:10.1007/BFb0085169.

722 Townsend, R. G., Solomon, S. S., Chen, S. C., Pietersen, A. N. J., Martin,
723 P. R., Solomon, S. G., & Gong, P. (2015). Emergence of complex wave
724 patterns in primate cerebral cortex. *The Journal of Neuroscience*, 35(11),
725 4657-4662. doi:10.1523/JNEUROSCI.4509-14.2015.

726 Xu, Y., Long, X., Feng, J., & Gong, P. (2023). Interacting spiral wave pat-
727 terns underlie complex brain dynamics and are related to cognitive pro-
728 cessing. *Nature Human Behaviour*, 7(7), 1196-1215. doi:10.1038/s41562-
729 023-01626-5.